Bidirectional Internship & Job Matching System Using Rule-Based Recommendation



Submitted By

EssaManzoor: 20692.

Abdul Hafeez: 19976.

Session 2021-2025

Supervised BY

_____

Supervisor Name

_____

A project proposal submitted to The IQRA National University, Peshawar in partial fulfillment of the requirements for the degree of

BACHELOR OF SCIENCE (BS) IN COMPUTER SCIENCE

DEPARTMENT OF COMPUTER SCIENCE
IQRA NATIONAL UNIVERSITY, PESHWAR-PAKISTAN
April, 2019

Table of Contents

1) Introduction || (Page 1)

2) Literature Review || (Page 1)

3) Problem Statement || (Page 1)

4) Objectives of Project || (Page 2)

5) Methodology for Implementation of Project || (Page 2)

6) Project Proposed Time Schedule || (Page 3)

7) Budget Description || (Page 3)

8) References || (Page 4)

9) Undertaking || (Page 4)

10) Supervisor Comments || (Page 5)

11) FYP Coordinator Remarks || (Page 6)

LIST OF FIGURES

Fig. No. **Fig. Description** **Page No.**

LIST OF TABLES

Table. No. **Table. Description** **Page No.**

LIST OF ACRONYMS

ACRONYMS	DESCRIPTION
UI	User interface
UX	User Experience
API	Artificial Programming Interface
DSS	Decision Support System
WWW	World Wide Web

INTRODUCTION

Finding suitable internships and entry-level jobs is a significant challenge for students, while companies struggle to identify candidates that match their requirements. Existing platforms mainly focus on one-way matching, which often leads to inefficiencies.

This project proposes a **Bidirectional Internship & Job Matching System** that provides recommendations to both students and companies using a transparent rule-based approach. The system is implemented as a mobile application using Flutter.

LITERATURE REVIEW

Various online job portals and internship platforms exist; however, most rely on manual searching or one-sided filtering. Recent studies emphasize decision support systems to improve matching efficiency. This project builds upon these concepts by introducing a two-way recommendation mechanism without relying on complex machine learning models.

PROBLEM STATEMENT

Current internship and job portals lack intelligent two-way matching mechanisms. Students apply to unsuitable positions, and companies receive a large volume of irrelevant applications, resulting in wasted time and poor matching outcomes.

OBJECTIVES OF PROJECT

- To develop a mobile-based internship and job matching system
- To provide bidirectional recommendations for students and companies
- To design a rule-based matching and scoring mechanism
- To create an easy-to-use Flutter-based interface

METHODOLOGY FOR IMPLEMENTATION OF PROJECT

The system uses a rule-based recommendation approach. Student and company profiles are compared based on predefined criteria such as skills, CGPA, location, and availability. A matching score is calculated and used to generate recommendation lists. Firebase/Supabase is used for backend services, authentication, and data storage.

1. Project Proposed Time Schedule

Sample is given below.

Activity	Month Year		Month Year	Month Year	Month Year	Month Year	Month Year	
Collection of Literature								
Study of Literature								
Analysis of Proposed Scheme								
Preparation of Schemes / Model								
Implementation of Schemes/Model								
Analysis & Simulation								
Result Formulation								
Final Write-up & Report Submission								

2. Budget Description

The project requires minimal budget as it is software-based. Development tools such as Flutter and Firebase are free to use. Testing will be performed using personal computers (PC) and mobile devices available to the students..

3. References

Firebase Documentaion,Google,2023
Doc.23, Doc 68.

Flutter Documetation,2023-2025.

4. Undertaking

.

We hereby declare that this project proposal is our original work and has not been submitted previously for any degree. All sources used have been properly acknowledged.

Signature of Students:

Essa Manzoor _____

Abdul Hafeez _____

5. Supervisor's Comments

Take recommendation of your supervisor for your project work here.

Signature of Supervisor:

_____

Name of Supervisor :

_____

Designation of Supervisor:

_____

6. Final Year Project Coordinator Remarks

Do not write anything here. For official use only.

Signature of Final Year Project Coordinator

_____

Designation

_____